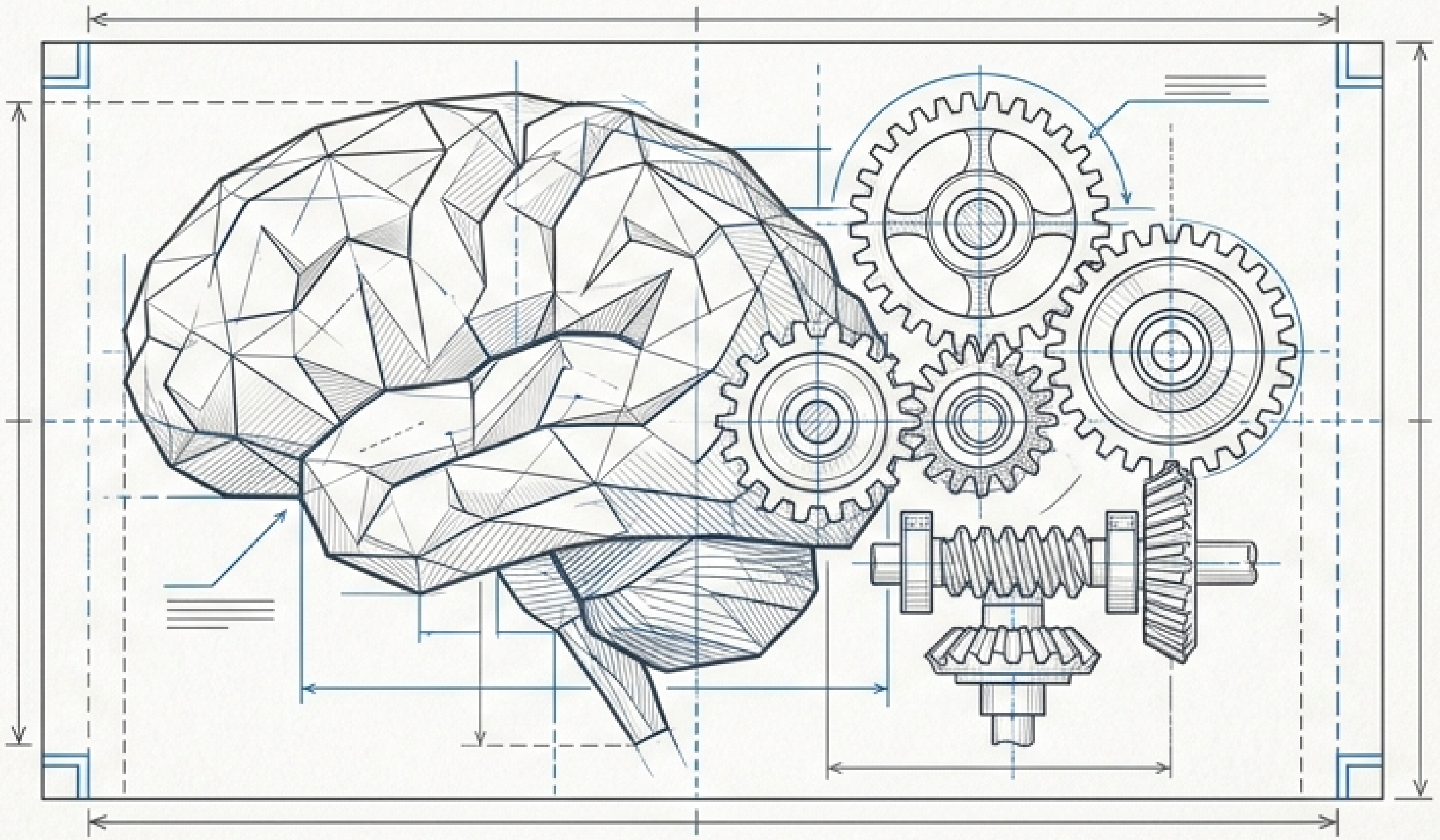
BUILD YOUR AI GAME: THE MASTER BLUEPRINT



A SYSTEMS ARCHITECTURE GUIDE FOR THE COMPLETE GAME LOOP

GAME ENGINE ARCHITECTURE: A CONTINUOUS HEARTBEAT

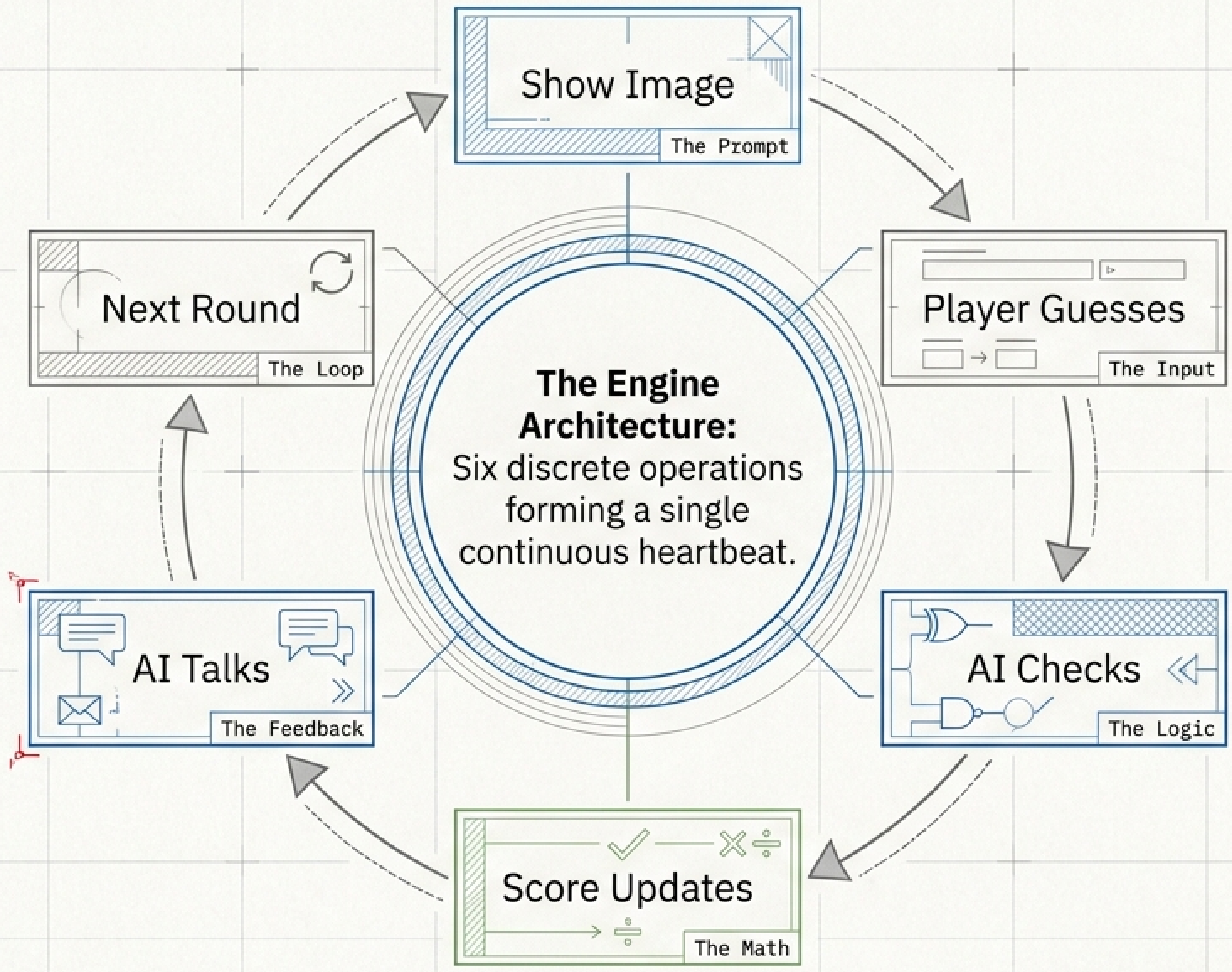
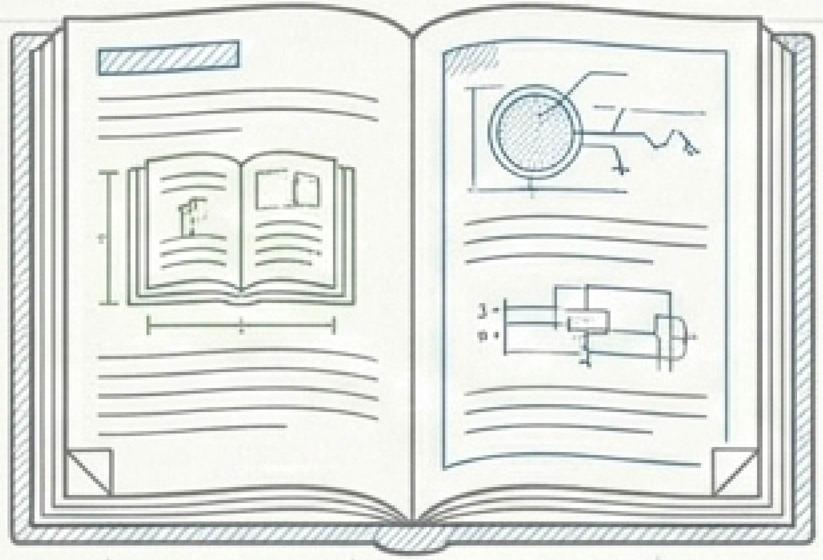


FIGURE 1.1 - CORE ENGINE CYCLE DIAGRAM

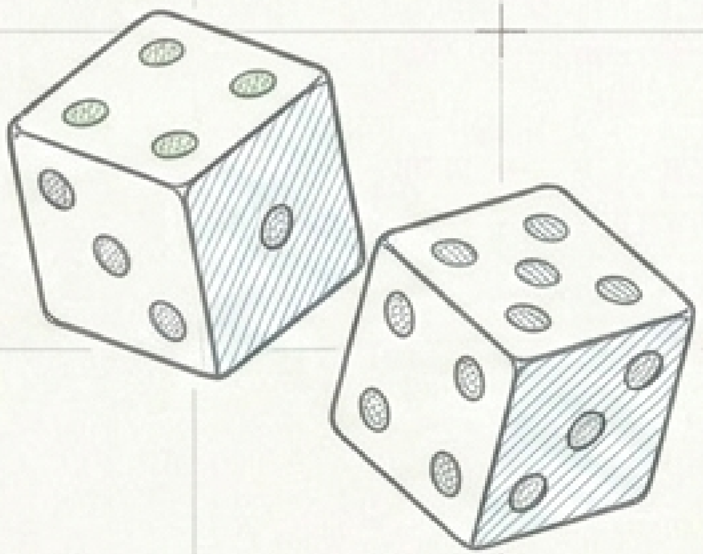
SYSTEM STATUS: OPERATIONAL

Developer's Inventory



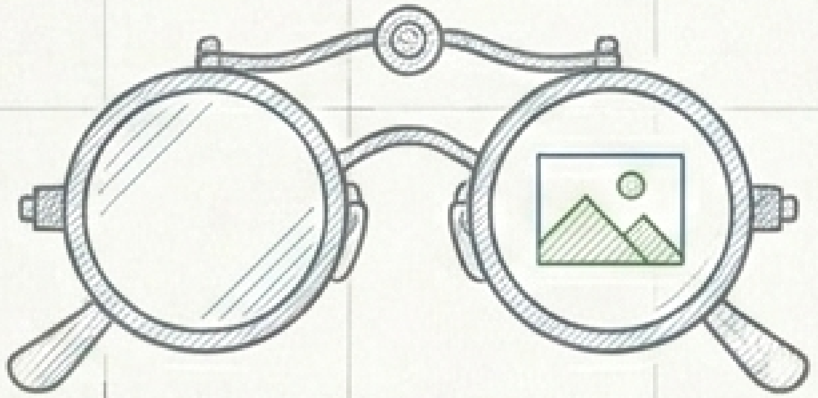
`import json`

Reads the data
(The Game's Memory).



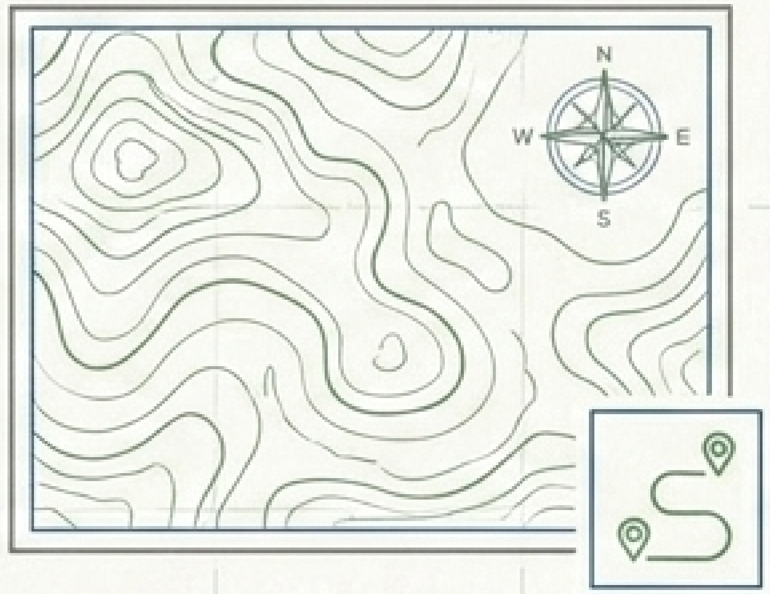
`import random`

Injects unpredictability
to keep the game fresh.



```
from PIL
import Image
```

Allows the engine to
physically render
visual assets.



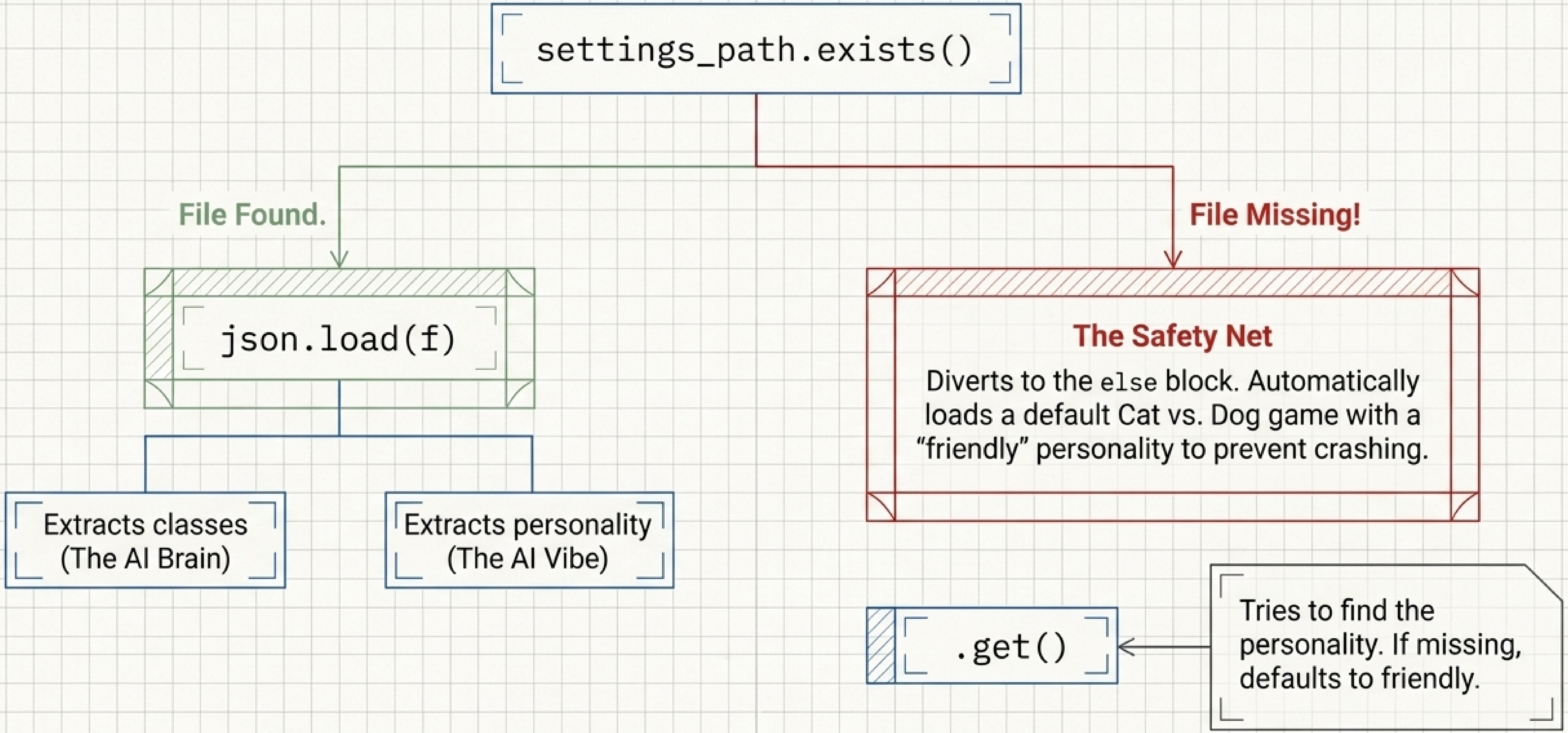

```
from pathlib
import Path
```

Navigates internal
folders to locate
hidden files.

FIGURE 1.2 - DEVELOPER'S INVENTORY

SYSTEM STATUS: OPERATIONAL

The Game's Memory



The Asset Pipeline

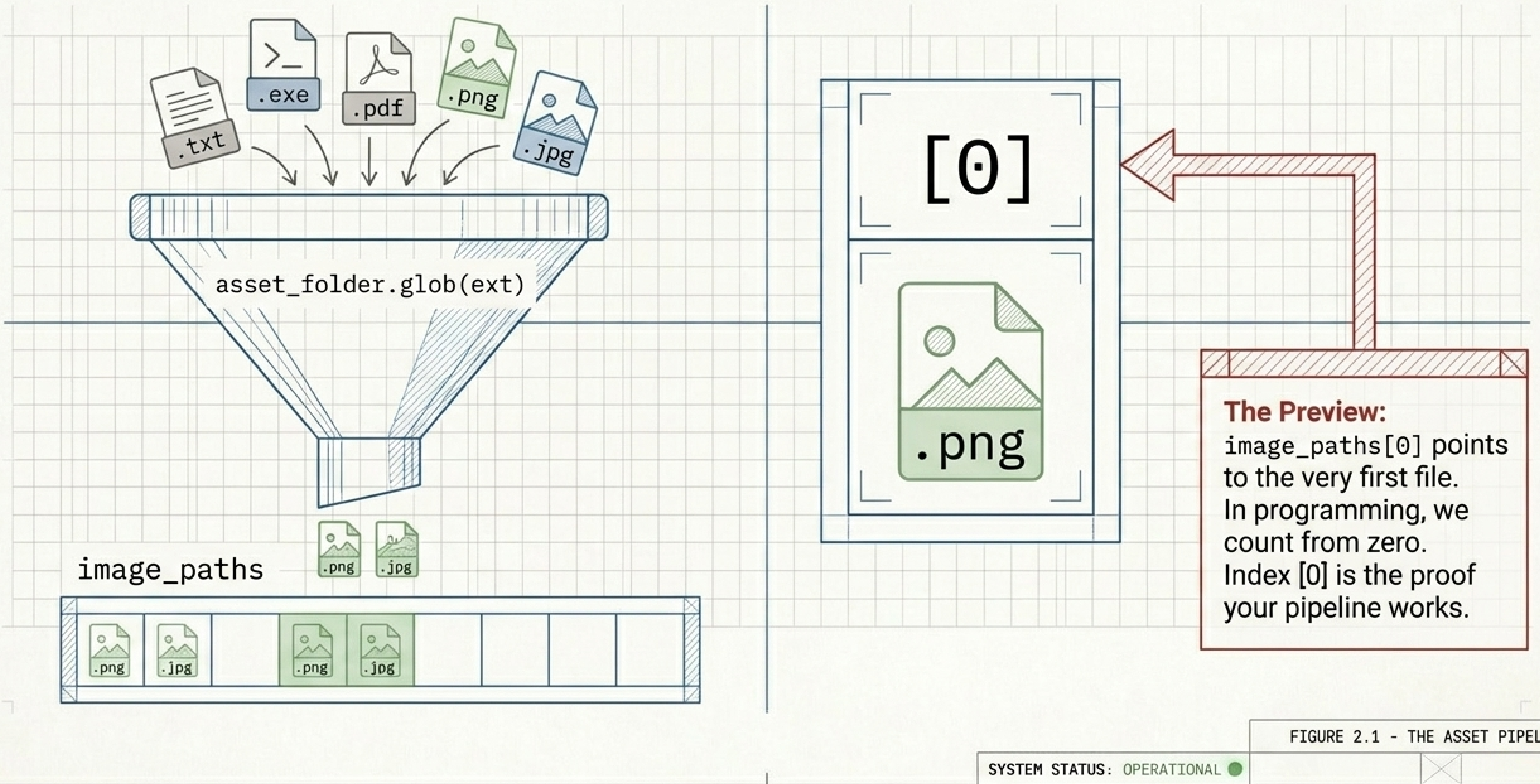


FIGURE 2.1 - THE ASSET PIPELINE

Game Skills Part I (Perception)

The Eyes (`show_image`)

```
plt.axis('off')
```

Strips away the mathematical X/Y rulers so the image looks like a game asset, not a math graph.

The Decoder (`answer_from_filename`)

```
name = image_path.stem.lower()
```

1 Strips away the '.jpg' to isolate the hidden answer.

2 This is a "Cheat Code" — guessing the answer based on text rather than actual visual intelligence.

FIGURE 3.1 - GAME SKILLS: PERCEPTION

Game Skills Part II (Personality)

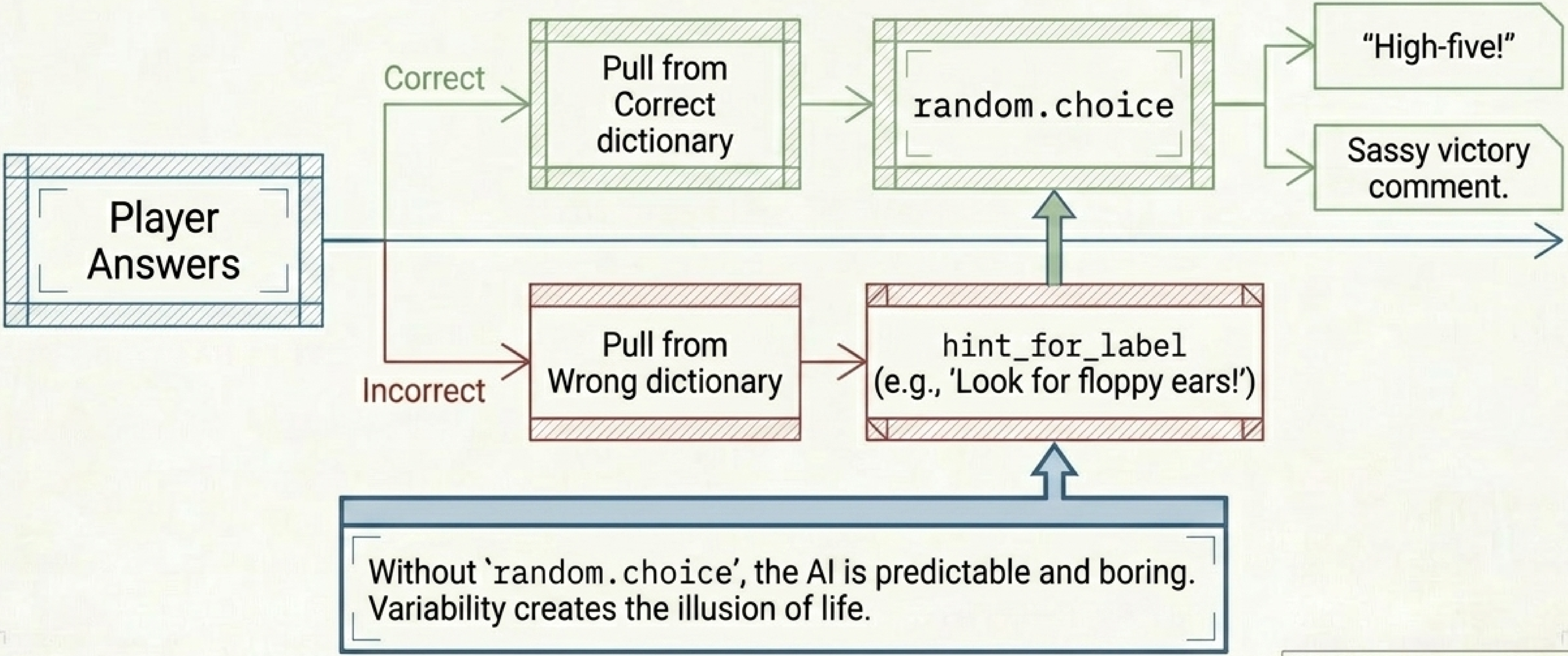


FIGURE 4.1 - GAME SKILLS: PERSONALITY

SYSTEM STATUS: OPERATIONAL

The Hand Test

The Dealer `random.choice(image_paths)`

Pulls one asset blindly.

The Prompt `input('Guess: ')`

Pauses the system, waiting for human interaction.

The Sanitizer "CAT" → "cat"

`player_guess.strip().lower()`

Cleans messy input data.

The Verdict Compare against `ai_answer`

Run it 3 times. If the AI changes its response, your RNG is working.

FIGURE 5.1 – THE HAND TEST

SYSTEM STATUS: OPERATIONAL

SYNTHESIS: The Game Loop (The Heartbeat)

⚠ The Indentation Hug:

Commands inside the bracket beat with the heart. Code outside the hug falls outside the outside the loop. If you un-indent the score, the game only calculates points once at the end!

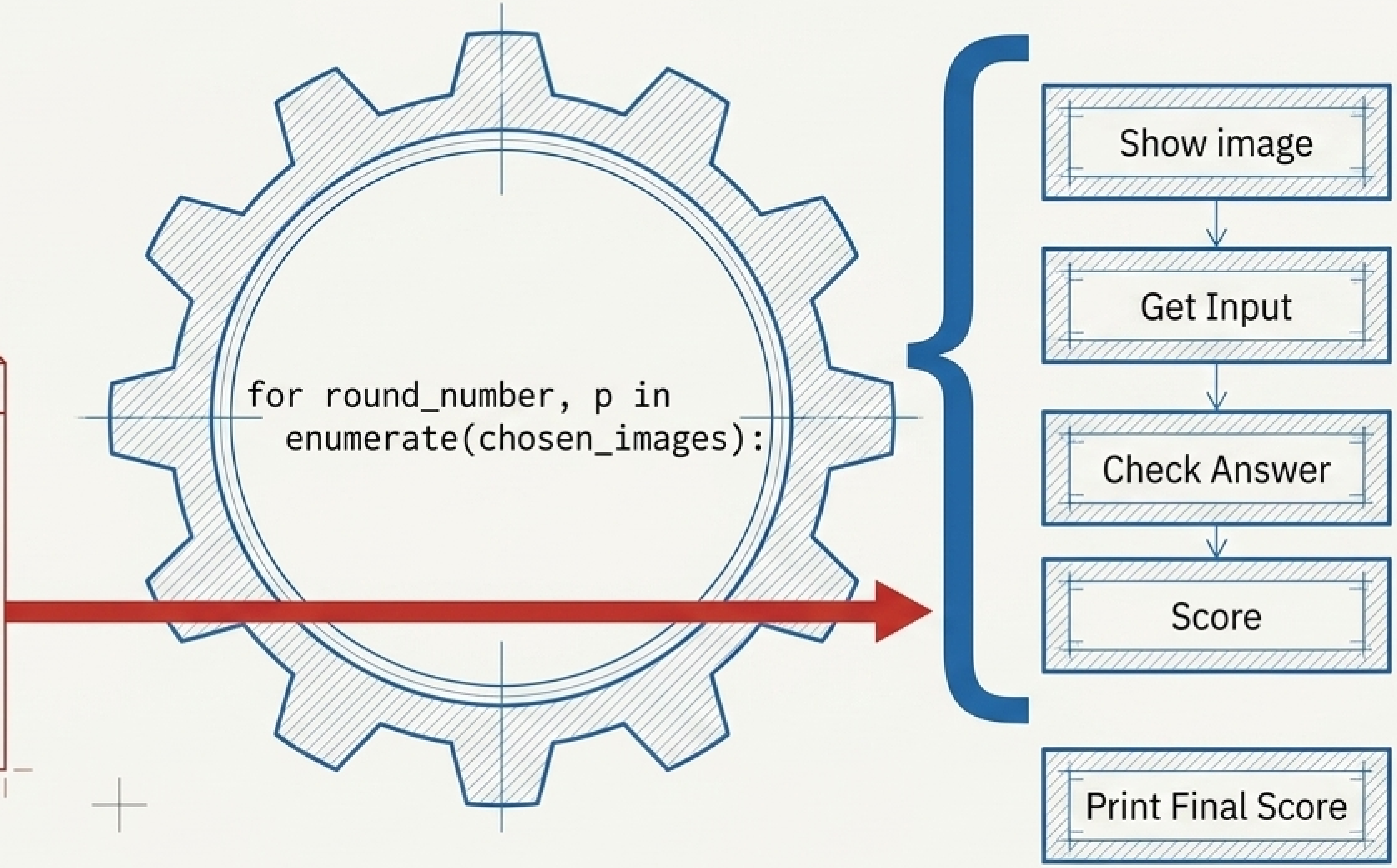
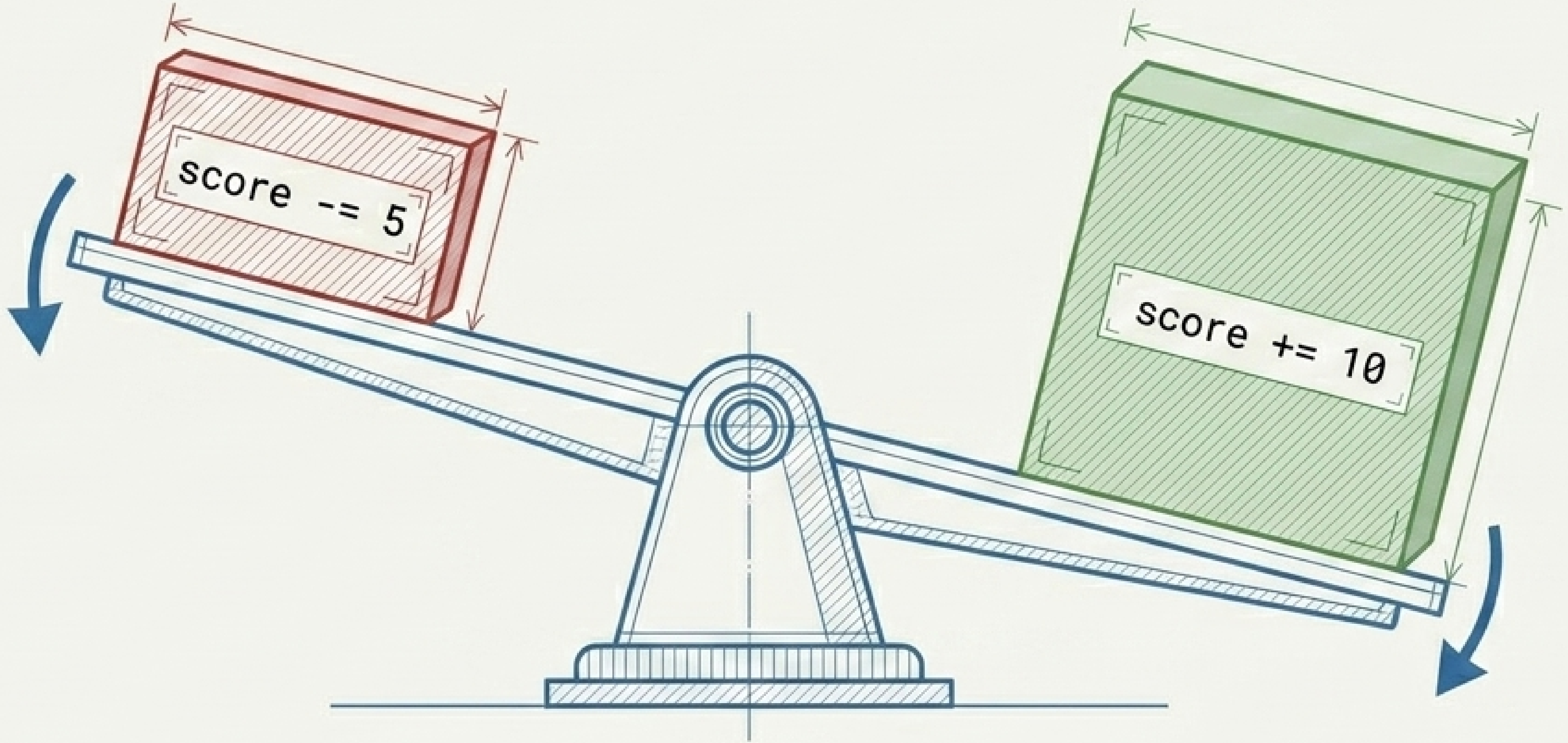


FIGURE 6.1 - THE GAME LOOP

SYSTEM STATUS: OPERATIONAL

Game Balance (Risk vs. Reward)

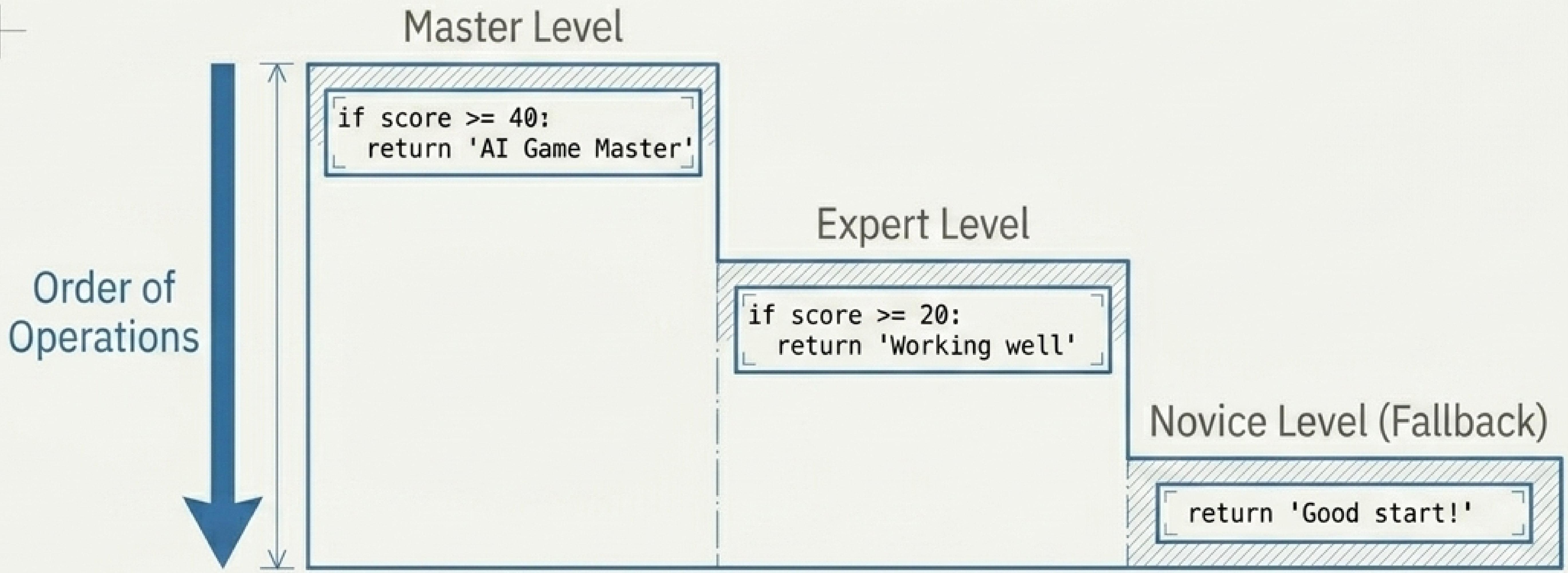


Game Balance Mechanics. You get a large reward for being right, but a calculated, smaller penalty for being wrong. If the penalty was -100, the game would be fundamentally broken and frustrating.

FIGURE 7.1 - GAME BALANCE

SYSTEM STATUS: OPERATIONAL

The Victory Screen



Top-to-Bottom Checking: Always check for the highest achievement first. If we put “Good Start” at the top of the code, everyone would trigger it immediately.

FIGURE 8.1 - THE VICTORY SCREEN | SYSTEM STATUS: OPERATIONAL

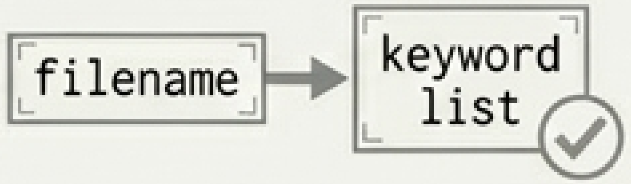
Cheat Code vs. Real AI

The Cheat Code

Method
`answer_from_filename`

The Input
Text Strings (cat_01.jpg)

The Mechanism
Keyword matching

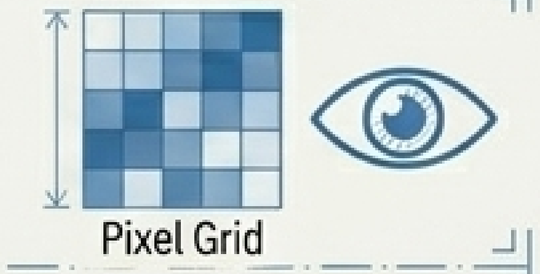


Verdict
Fast, rigid, and ultimately "cheating". The computer is blind to the image itself.

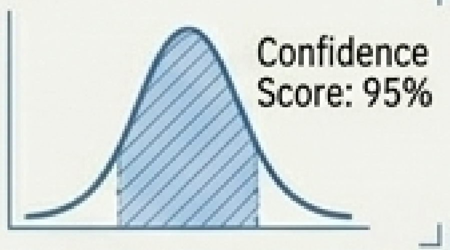
The Real Brain

Method
True AI Classifier

The Input
Raw Image Pixels



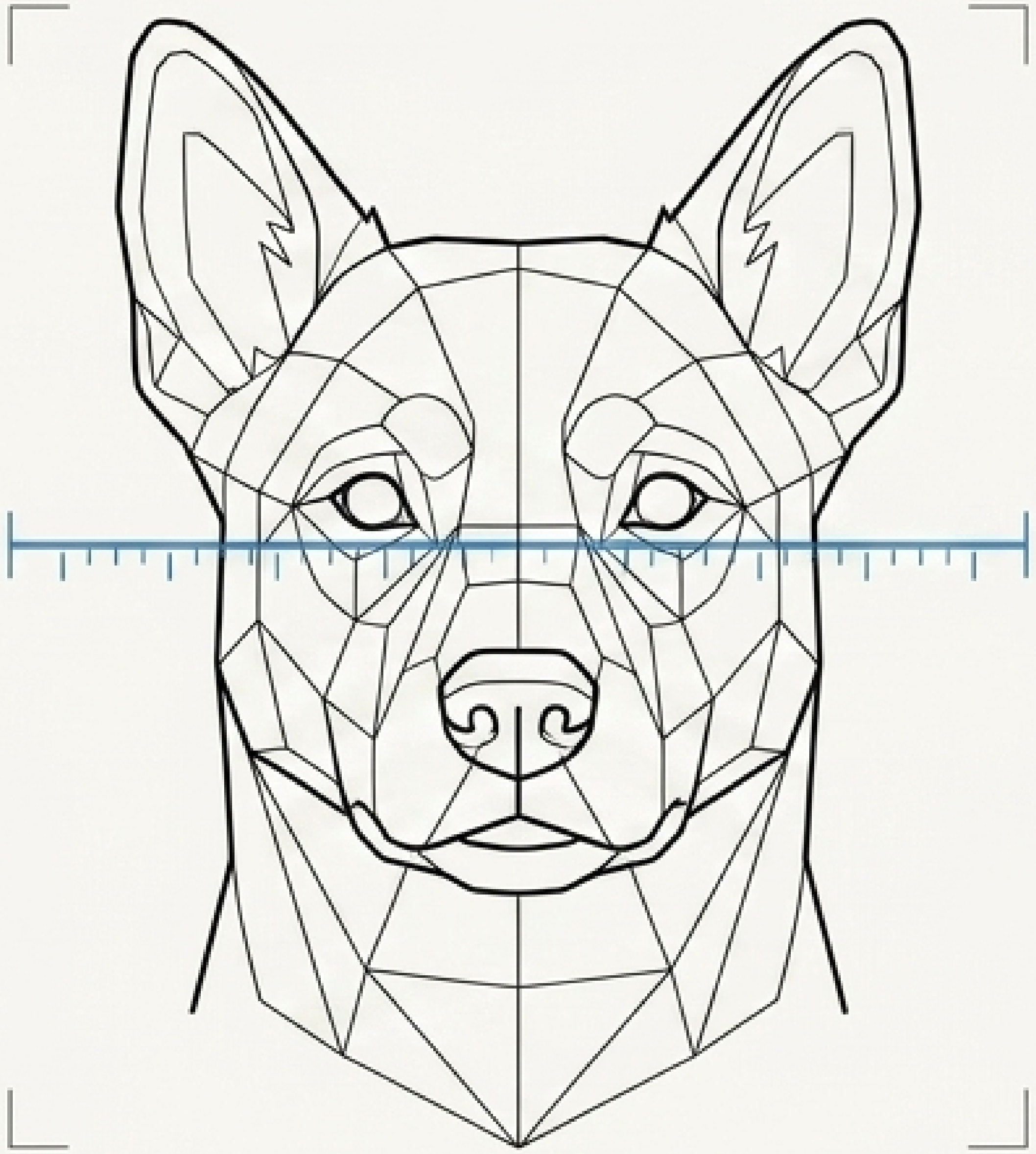
The Mechanism
Probability Confidence Scores

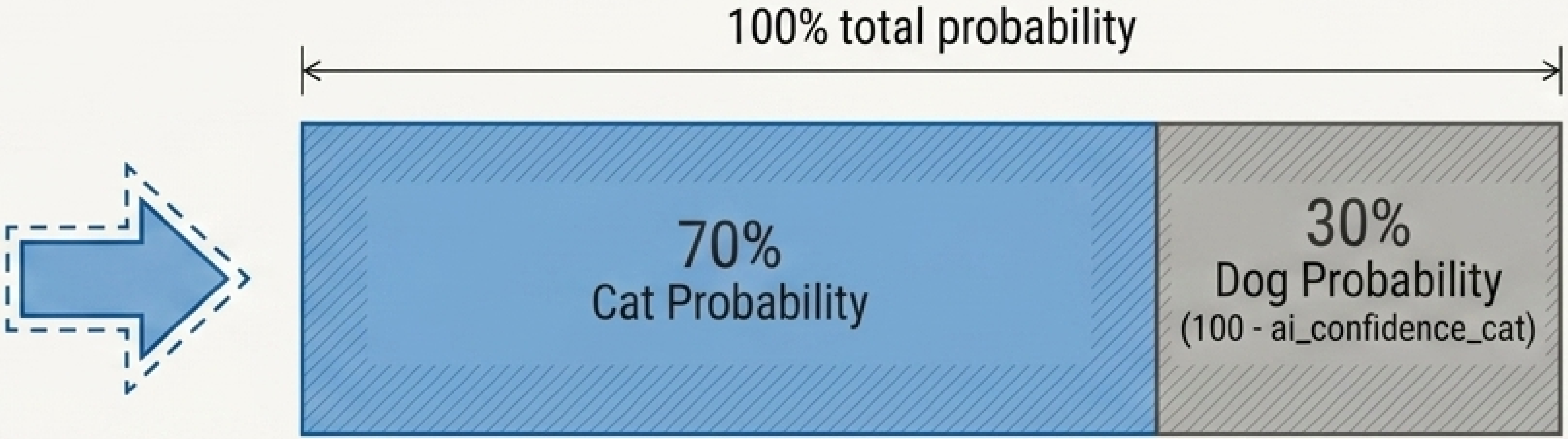


Verdict
Slower, but genuinely intelligent. Understands what it "sees" without relying on file labels.

FIGURE 9.1 - CHEAT CODE VS. REAL AI | SYSTEM STATUS: OPERATIONAL

The AI Vision Simulator





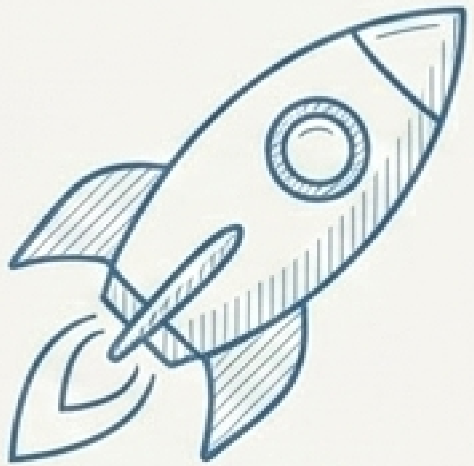
AI doesn’t think in absolutes; it thinks in Confidence Scores.
The machine analyzes features and outputs a zero-sum probability. Whichever bar is highest becomes its final declaration.

FIGURE 10.1 - THE AI VISION SIMULATOR | SYSTEM STATUS: OPERATIONAL

Hack Your Engine

The underlying architecture remains identical; only the data layer changes to create entirely new experiences.

Space Theme



Top Rank:
`Galactic Commander`
Bottom Rank:
`Space Cadet`

Cooking Theme

Top Rank:
`Five-Star Chef`
Bottom Rank:
`Dishwasher`

Zombie Theme

Top Rank:
`Ultimate Survivor`
Bottom Rank:
`Zombie Bait`

FIGURE 11.1 - HACK YOUR ENGINE | SYSTEM STATUS: OPERATIONAL

System Check

- **The Machine:** Loads assets, tracks the "`for`" loop heartbeat, manages score logic.
- **The Soul:** Injects personality via `random.choice` response and hint banks.
- LOADING **The Brain:** Swapping the filename "cheat code" for true pixel-vision classification.

System Online. You have successfully built a modular game architecture.
Prepare to install the AI vision engine.

FIGURE 12.1 - SYSTEM CHECK | SYSTEM STATUS: OPERATIONAL